

A Best Practices for Traveler Information Message (TIM) Deployment

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16. Abstract The following best practices guidance for the deployment of Traveler Information Messages (TIMs). This guidance is based on the results of the TIM Interoperability Testing conducted in Cheyenne, WY from October 7-10th was developed collaboratively within the Interoperability Technical Working Group (ITWG).				
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Table of Contents

1 Introduction	1
1.1 Document Overview	21
2 Best Practices for TIM Creation	3
2.1 Traveler Information Message (TIM) Development Guidance	3
2.2 TravelerDataFrame Implementation Guidance	6
2.3 GeographicPath Implementation Guidance	12
2.4 TravelerDataFrame.content Implementation Guidance	17
2.4.1 advisory	19
2.5 OffsetSystem Implementation Guidance	3328
2.6 NodeListXY Implementation Guidance	3328
2.7 GeometricProjection Implementation Guidance	3530
3 Additional Guidance	3933
3.1 Priority	3933
3.2 Geofence	4135
3.3 Timing	4236
4 Example TIM Messages	4438

List of Tables

Table 1 – Guidance for ASN.1 Entities contained within TravelerInformation	3
Table 2 – Guidance for ASN.1 Entities contained within TravelerDataFrame	6
Table 3 – Guidance for ASN.1 Entities Contained within GeographicPath	13
Table 4 – Guidance for ASN.1 Entities Contained within TravelerDataFrame.advisory	19
Table 5 Guidance for ASN.1 Entities Contained within OffsetSystem within the GeographicalPath dataframe	3328
Table 6 Guidance for ASN.1 Entities Contained within NodeSetXY in OffsetSystem within the GeographicalPath dataframe	3328
Table 7 Guidance for ASN.1 Entities Contained within GeometricProjection within the GeographicalPath dataframe	3530

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1 Introduction

As transportation systems evolve toward greater connectivity, the deployment of Vehicle-to-Everything (V2X) communications plays a pivotal role in enhancing roadway safety, efficiency, and situational awareness. Among the key components of V2X messaging are Traveler Information Messages (TIMs), which are vital for delivering real-time travel condition updates to drivers. These messages can include warnings about road closures, construction zones, adverse weather conditions, and other critical roadway events that impact driver decision-making and traffic flow.

This document provides best practices for the creation, deployment, and interpretation of TIMs to support consistent and interoperable V2X communications across jurisdictions and platforms. By establishing these best practices, transportation agencies and technology providers can ensure that TIMs are reliably understood by all vehicles equipped with V2X capabilities, regardless of manufacturer or location.

The recommendations presented in this document are grounded in established standards, including the Society of Automotive Engineers (SAE) J2735¹ and J2540² standards, which define the structure and content of TIMS using ASN.1 encoding and the list of International Traveler Information Systems (ITIS) codes and phrases used to display messages to drivers, respectively. Additionally, the Manual on Uniform Traffic Control Devices for Streets and Highways (MUTCD)³ is referenced to align TIM content with conventional traffic control practices and ensure that digital traffic messages and signs mirror the clarity and intent of physical signage.

Insights from a TIM Interoperability Testing event held October 7 – 10, 2024 near Cheyenne, Wyoming further inform these best practices. This event brought together stakeholders from across the transportation and technology sector to evaluate the current state of TIM deployment and identify opportunities for improvement. The findings underscore the need for a more structured and harmonized approach to TIM creation, particularly as V2X technology moves toward nationwide implementation.

By following the recommendations in this document, stakeholders can contribute to a safer, and more connected transportation system.

¹ https://www.sae.org/standards/content/j2735_202409/

² https://www.sae.org/standards/content/j2540_201906/

³ MUTCD 11th Edition: https://mutcd.fhwa.dot.gov/kno_11th_Edition.htm

1.1 Document Overview

Sections included in this Best Practice Guide include:

- **Section 1: Introduction** – Provides an overview of the purpose of these best practices and outlines the sections.
- **Section 2: Best Practices for TIM Creation and Interpretation** – Provides tables of the different data frames and elements within TIM messages and guidance on how to set and interpret the values.
- **Section 3: Additional Guidance** – Provides additional best practices on how to determine message priority, and TIM geofence and timing settings
- **Section 4: Example TIM Messages** – Provides examples of various TIMs that follow the guidance provided in this guide.

2 Best Practices for TIM Creation

2.1 Traveler Information Message (TIM) Development Guidance

This section defines the contents of the J2735 TravelerInformation ASN.1 entity, often referred to as TIM. The Message ID for TIMs is set to 10 per SAE J2735. Best practices on how to create and interpret the fields within the TIM are included in the tables below.

NOTE: Unless marked as Do Not Include, all SAE J3735 OPTIONAL items must be included as set per the Requirement for TIM Creation column.

Table 1 – Guidance for ASN.1 Entities contained within TravelerInformation

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
msgCnt	MsgCount	M	- When creating a new TIM, always set msgCnt to 0. - Each time there is a content update, set the new msgCnt value to msgCnt + 1. - This Value is modulo 127, so msgCnt should be set to 0 for the update following a msgCnt value of 127. - If there are two TIMs with the same packetID, and one has a msgCnt = 120+ and another msgCnt = 0-10, a roll over may have occurred. Verify by comparing the timeStamp field.	- When messages have the same packetID, the msgCnt can be used to identify the most recent TIM. The TIM with the highest msgCnt is usually the most recent message - If there are two TIMs with the same packetID, and one has a msgCnt value of 100+ and the other has a msgCnt = 0-10, then the timeStamp should be compared to identify the most recent TIM. The latest timeStamp is the most recent message.

Best Practices for TIM Creation

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
timeStamp	MinuteOfTheYear	O	• Mandatory. The ITWG recommends that this field is mandatory even though is optional according to J2735. • Set the timestamp to the time that the TIM is generated. ◦ Include the number of elapsed minutes of the current year (from 0 up to 527039) based on UTC time. The value 527040 is used for invalid. • This field should be updated each time the msgCnt is updated.	• For TIMs that have the same packetID and the msgCnt values are near the turnover (from 127 to 0), compare the timestamp to identify the latest TIM.

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
packetID	UniqueMSGID	O	• Mandatory • Set to a relatively unique ID within the context of the deployment. • Generate in the format of a 9-octet string. ◦ The first 3 bytes will be used to define the GNIS code for the deployment area • This value should NOT change when the content of the TIM is updated.	• Identify unique messages based on their packetID • If there a multiple TIMs with the same packetID, use msgCnt and timeStamp to identify updates to TIMs
urlB	URL-Base	O	• Do not include.	• Ignore

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
dataFrames	TravelerDataFrameList	M	- Sequence of up to eight instances of TravelerDataFrame. - At least 1 TravelerDataFrame should be included. - If multiple TravelerDataFrames are used, then they must be dependent on each other. - Example: In variable speed limit messages, a deployer may decide to generate traveler data frames with the same packet ID for each of the speeds. - Refer to Table 2Table 2.	If there a multiple TraverlaerDataFrames, the TIMs relate to each other and should have the same packetID
regional	RegionalExtension	O	Do not include.	Ignore

2.2 TravelerDataFrame Implementation Guidance

This section provides guidance on the creation of the TravelerDataFrame which is used to send a single message within a TIM. The traveler data frame provides information on the type of sign and content within the TIM as well as the time and priority of the message. Note that there can be multiple data frames within a TIM. All times based on UTC.

Table 2 – Guidance for ASN.1 Entities contained within TravelerDataFrame

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	Requirement for TIM Creation	TIM Interpretation
doNotUse1 formerly: sspTimRights	SSPindex	M	Set to zero. This is a legacy field maintained for backwards compatibility.	Ignore
frameType	TravelerInfoType	M	- Set this element to roadSignage (2) if a state or local deployment agency is generating the message. - Set this element to commercialSignage (3) if a commercial entity is generating this message. - Do not use advisory (1).	Identify if a state or local agency or a commercial entity is creating and broadcasting this message based on frameType value.
msgId		M	- Choice between furtherInfoID and roadSignID. - It is recommended to only use the furtherInfoID field.	
furtherInfoID	FurtherInfoID		- Mandatory - Set to zero-0000	
roadSignID	RoadSignID		Do not include. There can be confusion with setting multiple heading and position elements when creating the roadSignID.	Ignore

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	Requirement for TIM Creation	TIM Interpretation
startYear	DYear	O	• Mandatory • Set to the integer value of the year in the Gregorian calendar that the TIM begins broadcasting. • Valid values: 2000 up to 4095. • Ex: set to 2025 for the year 2025. • StartYear is needed to identify the year that the MinuteOfTheYear entity applies to. It adds clarity in the case of year roll-overs or time restricted messages (i.e. work zone TIMs).	• This value is the year the TIM is generated. Look to startTime or durationTime for further details on TIM validity • If durationTime = 32000, startYear is not important if in the past

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	Requirement for TIM Creation	TIM Interpretation
startTime	MinuteOfTheYear	M	- Set to minute within the startYear that the sign validity starts. - Valid Integer values: 1 to 527039. 527040 is used for invalid.	- If startYear is the current year, use the startTime value to know how far into the year the message starts. Used to determine if the TIM is currently active. - If durationTime = 32000 and the startYear is in the past, startTime is not important
durationTime	MinutesDuration	M	- Set to the duration of the sign validity in minutes (up to 22.2 days). - Valid values: 1 to 32000. Use value 32000 for signs that last forever.	- A value of 32000 means the TIM is indefinite until deleted. - Use duration time to determine if the TIM is still valid or applicable.

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	Requirement for TIM Creation	TIM Interpretation
priority	SignPriority	M	- Set the relative importance of the message display on a scale from 0 (least important) to 7 (most important) when needing to display multiple signs. 0 is used if the IOO doesn't want to rank messages 7 is reserved for most critical signs. Example: Wrong way driver alert	- This value is based on the recommendation from the IOO of the importance of message when displaying multiple messages - Ultimately up to consumer to decide display priorities - The message with the highest priority should be displayed and considered first.
doNotUse2 formerly: sspLocationRights	SSPindex	M	Legacy field - set to zero.	Ignore
regions	GeographicalPath	M	- Can have a sequence of up to 16 instances of geographical paths. - Refer to Table 3 and Table 4.	
doNotUse3 formerly: sspMsgTypes	SSPindex	M	Legacy field - set to zero.	Ignore
doNotUse4 sspMsgContent	SSPindex	M	Legacy field - set to zero and ignore upon reception	Ignore

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	Requirement for TIM Creation	TIM Interpretation
content		M	- Select one of the five content types listed below - Use ITIS codes from a defined message set like in SAE J2540	- Identify the content type of the TIM based on this field. - Interpret ITIS codes and text for further details needed to comprehend the message
advisory	ITIScodesAndText		Refer to Table 4 Error! Reference source not found.	Interpret ITIS codes and text for further details needed to comprehend the message
workZone	WorkZone		Do not use. Use the advisory content type.	Interpret ITIS codes and text for further details needed to comprehend the message
genericSign	GenericSignage		Do not use. Use the advisory content type.	Interpret ITIS codes and text for further details needed to comprehend the message
speedLimit	SpeedLimit		Do not use. Use the advisory content type.	Interpret ITIS codes and text for further details needed to comprehend the message

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	Requirement for TIM Creation	TIM Interpretation
exitService	ExitService		Do not use. Use the advisory content type.	Interpret ITIS codes and text for further details needed to comprehend the message
url	URL-Short	O	Do not include. Use the advisory content type.	Interpret ITIS codes and text for further details needed to comprehend the message
contentNew	TravelerDataFrameNewPartIIIContent	MO	Set to zero	Ignore

2.3 GeographicPath Implementation Guidance

GeographicalPath is used to define a geographical area that the TIM is applicable for. It is used in many V2X message types. The geographical path can be defined using a path, a circle, or a polygon. A summary of how to distinguish between the different geofence types and which fields to use for setting the relevant direction is included below. Further details for setting the appropriate geographic path are listed in Table 4.

Geofence Types:

- “Path” – path(OffsetSystem) with closedPath = False
 - Use for single road TIMs
 - Use TravelerDataFrame.regions.directionality(DirectionOfUse)
- “Polygon” – path(OffsetSystem) with closedPath = True
 - Use for messages with multiple roads
 - Complicated road structures like cloverleafs
 - Use TravelerDataFrame.regions.direction(HeadingSlice) if there is a heading restriction
- “Circle” – geometry (GeometricProjection)
 - Do not use TravelerDataFrame.regions.direction(HeadingSlice) or TravelerDataFrame.regions.directionality(DirectionOfUse)

- Use `TravelerDataFrame.regions.description.geometry.direction(HeadingSlice)` in `GeometricProjection`

Table 3 – Guidance for ASN.1 Entities Contained within GeographicPath

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
name	DescriptiveName	O	Do not include.	Ignore
Id	RoadSegmentReferenceID	O	Do not include.	Ignore
region	RoadRegulatorID	O	Do not include.	Ignore
id	RoadSegmentID	O	Do not include.	Ignore
anchor	Position3D	O	- This field is mandatory if using a “linear path” based geofence with offsets - Anchor should be set 10 meters before first node in path - If anchor is used, the elevation field is mandatory	Use this field to decode “linear path” geometries that use relative offset nodes.
lat	Latitude		- Set latitude of the TIM starting point - Valid values: -900000000 – 900000000 900000001 is used when the latitude is unavailable.	Use latitude value to in conjunction with longitude and elevation to determine location of the anchor needed for the path geofence

Best Practices for TIM Creation

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
Long	Longitude		- Set longitude of the anchor needed for the path geofence - Valid values: -1799999999 – 1800000000 1800000001 is used when the longitude is unavailable.	Use longitude value to in conjunction with latitude and elevation to determine location of the anchor needed for the path geofence
elevation	Elevation		- Valid values: -4095 to 61439 meters - Use value -4096 when the elevation is unknown and/or not required for the specific use case. - If there are inclines or declines in the road, it can be useful to define the path or area in a 3-D space. or in the case of identifying location for over-head ramps and tunnels.	Use elevation value to in conjunction with longitude and latitude to determine location of the anchor needed for the path geofence
regional	RegionalExtension		Do not include.	Ignore

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
laneWidth	LaneWidth	O	- Only include if using a path geofence. The circle geometric projection has its own laneWidth field. - Set to the lane width in centimeters. This value should be consistent with the nodes created in the path field such that the nodes are the center of the specified lane width. If a single lane is being mapped, set to the single lane width. If multiple lanes are being mapped, then set to the width of the combination of those lanes. - Valid values: 1 – 32767 meters. The Maximum allowed lane width is 327.67 meters.	The TIM is valid for the lane width defined along a path by this value
directionality	DirectionOfUse	O	- Set to 1 for forward direction, 2 for reverse, or 3 for both depending on the direction of travel that the TIM is applicable for - Mandatory if using a “path” geofence with closedPath = FALSE	There should be a heading angle defining travel direction applied around each path segment direction

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
closedPath		O	- Mandatory if using a “path” data element - Set to FALSE if creating a “Path” geofence - Set to TRUE if creating the “Polygon” geofence	- If set to false, the region for a TIM is along a path - If set to TRUE, the region for the TIM is defined by a polygon
direction	HeadingSlice	O	Only use for “polygon” geofence if there is directionality within the polygon	
description		O	- Set to one of the fields below an option below: - path – OffsetSystem, (used to create paths and polygon geofences) - geometry – GeometricProjection (used to create a circle geofence) - Do not use oldRegion	- path – used for “path” or “polygon” geofence. Distinguish between the two based on the closedPath field. - geometry – used for a “circle” geofence - Ignore oldRegion

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation
path	OffsetSystem	O	• M if using “path” or “polygon” geofence • Node points happen in order from first node to the last node following a roadway • Use ll offsets for longer distance paths and xy offsets can for smaller distance paths • Must use a road geometry definition with a high enough density of node points to maintain the path on the roadway • Refer to Error! Reference source not found, Table 10	• Use to define a path or polygon geofence that the TIM is applicable for
geometry	GeometricProjection	O	• Mandatory if using a “circle” geofence • Need large enough radius for circle geofence. • Refer to Table 7	• Use to determine the circle geofence that the TIM is applicable for
oldRegion	ValidRegion	O	• Do not use – legacy method.	• Ignore
regional	RegionalExtension	O	• Do not include.	• Ignore

2.4 TravelerDataFrame.content Implementation Guidance

There are five content types for TIMs: **advisory**, **work zone**, **generic**, **speed limit**, and **exit signs**. Each content type enables deployers to input ITIS codes and text phrases, in accordance with the J2540 standard, to define messages intended for display to travelers. To streamline the deployment process and reduce complexity in selecting the appropriate content type, these categories have been consolidated into a single default type. The **advisory** content type has been designated as the default due to its larger text field capacity, allowing for more comprehensive messaging.

To differentiate non-regulatory messages/signs from regulatory restriction signs, the ITIS code and phrase for “advice” is used. It is important to note that some signs may convey identical messages but differ in color to indicate context-specific usage. For example, orange signs are typically used for work zones and temporary traffic control signs while pink signs are used for incident management scenarios. For further details on sign color conventions, refer to **Section 1D.05** in the MUTCD (11th Edition).

2.4.1 Advanced Warning Messaging

For many signs, deployers may want to provide advance notice of upcoming travel conditions. This is usually indicated by appending the ITIS phrase and code for “ahead” to the message. When possible, it is recommended to include the specific distance that the incident occurs (e.g. “2 miles ahead”). If the exact distance is unknown, the term “ahead” alone is acceptable.

Since dynamically calculating distance can be challenging, it is recommended that the geofence associated with the TIM message include the distance parameters. This enables OBU manufacturers to compute and display accurate distance information. Refer to the MUTCD (11th Edition) **Section 6B.04** for additional guidance on advance warning sign placement:

- Typical distances for placement of advance warning signs on **freeways and expressways** should be longer because drivers are conditioned to uninterrupted flow. Therefore, the advance warning sign placement should extend on these facilities as far as ½ mile or more.
- On **urban streets**, the effective placement of the nearest warning sign to the TTC zone, in feet, should range from 4 to 8 times the speed limit in mph, with the high end of the range being used when speeds are relatively high. When two or more advance warning signs are used on higher-speed streets, such as major arterials, the advance warning area should extend a greater distance (see Table 6B-1).
- Since **rural highways** are normally characterized by higher speeds, the effective placement of the first warning sign in feet should be substantially longer—from 8 to 12 times the speed limit in mph. Since two or more advance warning signs are normally used for these conditions, the advance warning area should extend 1,500 feet or more for open highway conditions

The geofence for the advance warning message should terminate precisely where the main TIM geofence begins.

2.4.2 Speed Limit Messaging

To define numeric speed limits, it is recommended to use the small number ITIS codes defined in **J2540 section 6.35**. Commonly used values include:

- n15 – n85 are commonly for speed limits
- ITIS code 8720 for miles per hour (mph)
- ITIS code – 8721 for kilometers per hour (kph)

2.4.3 Truck Parking Availability

The **MUTCD (11th Edition) Section 2I.15** on Signing for Truck Parking availability provides the following guidance:

- Truck Parking Availability signs should be located **3 to 5 miles** in advance of the nearest parking facility.
- The parking facilities displayed on the sign should be **no more than 60 miles** from the sign location.

Note: These best practices do not currently support commercial signage for exit services. For additional guidance on MUTCD supported exit service signs, please refer to:

- **section 2C.12** for exit and ramp speed signs
- **section 2E.53** for next exit signs
- **section 2E.54** for weigh station signing
- **section 2F** for toll road signs.

2.4.4 Forming ITIS Phrases for TIMs

The Federal Highway Administration (FHWA) provides a structured approach for Portable Changeable Message Signs (PCMS) in its PCMS handbook, which can be adapted for ITIS phrases construction:

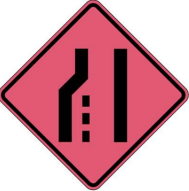
- 1) Describe the problem or situation
Example: "Fog ahead", "Lane closed"
- 2) Identify the location or distance ahead
Example: "2 miles ahead", "Use exit 12"
- 3) Provide motorist instruction or recommended action
Example: "Merge left", "Reduce speed"
- 4) (Optional) Include impact or effect on travel
Example: "Expect delay"

The table below lists examples of traveler messages and their corresponding MUTCD signs that will be displayed to On-board units (OBUs) in vehicles.

2.4.5 advisory

Table 4 – Guidance for ASN.1 Entities Contained within TravelerDataFrame.advisory

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
itis	ITIScodes	Use applicable ITIS codes and text:	Images are associated with the ITIS codes in the TIM creation column

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		- closed-to-traffic, local-traffic, only 769, 9478, 7747	
		- Incident, ahead 531, 13569	
		- accident, ahead 513, 13569	
		- left-lane, blocked-ahead 8195, 776	
		- left, <small number>, lanes, blocked-ahead 13580, <small number>, 13588, 776	
		- right-lane, blocked-ahead 8196, 776	
		- right, <small number>, lanes, blocked-ahead 13579, <small number>, 13588, 776	

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		• <wind type>, may-exceed, <number>, mph • 5131, 7759, <small>, 8720 • Sample wind type options: ○ hurricane-force-winds: 5123 ○ Tropical-storm 5124 ○ Storm-force-winds: 5126 ○ Strong-winds: 5127 ○ Gusty-winds: 5131 • strong-winds, high-profile-vehicles, hazardous-driving-conditions • 5127, 9233, 6405 • Caution, <wind type> • 12330, <wind type> Regulatory high profile vehicle restrictions: • strong-winds, truck-restrictions, no-high-profile-vehicles, below, gross-weight-limit, <number>, pounds • 5127, 2563, 2569, 7682, 2577, <number>, 8739	   

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		- truck-restriction, gross-weight-limit, <number>, pounds/tons 2563, 2577, 8739/8740	 
		- truck-restriction, length-limit, <number>, feet 2563, 2577, <number>, 8710	
		- wrong-way 12310	
		- Danger, ahead, vehicle-traveling-wrong-way, confirmed-report 6917, 13569, 1793, 2822	
		- hurricane, evacuation-route 5122, 8468	

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		- visibility-reduced 5383 Note: Any geofence type can be used in conjunction with a visibility reduced message/sign	
		- dense-fog, ahead 5377, 13569 Note: A path geofence should be used for the dense fog ahead message/sign	
		- be-prepared-to-stop 7201	
		- Road-construction, be-prepared-to-stop 1025, 7201	
		- traffic-congestion, ahead 263, 1356996	

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		- Road-construction, traffic-congestion, ahead 1025, 263, 72013569	
		- left-turn/right-turn, advice, on-curve, speed-limit, <small-number>, mph 13594/13599, 7712, 8026, 268, <small-number>, 8720	 
		- hairpin-left/hairpin-right, advice, on-curve, speed-limit, <small-number>, mph 13618/13617, 7712, 8026, 268, <small-number>, 8720	 
		- winding-road-to-left/winding-road-to-right, advice, on-curve, speed-limit, <small-number>, mph 13641/13640, 7712, 8026, 268, <small-number>, 8720	 

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		• debris-on-roadway • 1284 • Slippery, wet-pavement • 5897, 5895 • Surface-water-hazard • 5893 • Ice • 5906	

Best Practices for TIM Creation

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		- severe-weather 4865	
		- blowing-snow 5385	
		- flooding, ahead 3073, 13569	
		- flooding, road, closed-ahead 1301, 11793, 771	/
		- emergency-vehicles-on-roadway 1796	/
		- emergency-vehicles-at-scene 6925	
		- Road-construction 1025 Optional: Ahead - 13569	

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Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		- look-out-for-workers 6952 - do-not-enter 12314 - Stop 12294 - Yield – 12295	

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Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		Advisory speed limit: - Advice, speed limit, <small number>, mph/kmph 7712, 268, <small number>, 8720/8721 Regulatory speed limit: - speed limit, <small number>, mph/kmph 268, <small number>, 8720/8721 - Speed limit, reduced, <small number>, mph/kmph, ahead 268, 12302, <small number>, 8720/8721, 13569 - Speed limit, trucks, buses, n45, mph 268, 9227, 9228, 12589, 8720 - Vehicle groups are listed in section 6.50 in J2540	   



STOP AHEAD
PAY TOLL

LAST EXIT
BEFORE TOLL

RUNAWAY
TRUCK RAMP
1 MILE ↗

RUNAWAY
TRUCK RAMP
↖ 1 MILE

Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		- truck, parking, ahead 9227, 4120, 13569	
		- Gasoline/diesel, ahead 11016/11015, 13569	
		- Test-your-brakes, parking-area, ahead 7174, 11791, 13569	
		- Snow-chains-required, parking-area, ahead 6148, 11791, 13569	
		- Snow-tires-or-chains-required, all-vehicles 6156, 9217	

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Name	ASN.1 Entity (J2735)	TIM Creation	TIM MUTCD Image Interpretation
		• Weigh station, <number>+ mile, ahead • 11792, <number>, 8711, 13569 • Weigh station, open, next, right • 11792, 892, 13582, 13579	
text	ITIS text	Used to express proper nouns such as local expressions that do not appear in ITIS tables like street names.	

2.5 OffsetSystem Implementation Guidance

The OffsetSystem field is used to define the nodes and scale for the setting the path and polygon geofence types.

Table 5 Guidance for ASN.1 Entities Contained within OffsetSystem within the GeographicalPath dataframe

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	Requirement (Use)
scale	Zoom	O	Set the scaling factor for the region, or offset points within a map in unites of 2^N. N: 0 = 1:1 zoom (no zoom) 1 = 2:1 zoom 2 = 4:1 zoom, etc. Valid values: 0 up to 15
offset			
xy	NodeListXY		Set these fields for most paths that are not long distance. See Table 6Table 11 .
ll	NodeListLL		Only use for long distance paths

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2.6 NodeListXY Implementation Guidance

The NodeListXY data frame is a choice of NodeSetXY or ComputedLane. Computed lane is generally used when there are adjacent lanes that are the same width at an intersection. The offset will be calculated from a reference lane. Guidance for creating MAP messages listed in CTI 4501 was incorporated into this section.

Table 6 Guidance for ASN.1 Entities Contained within NodeSetXY in OffsetSystem within the GeographicalPath dataframe

Name	ASN.1 Entity (J2735)	Mandatory (M)/ Optional (O) according to SAE J2735	TIM Creation	TIM Interpretation

Best Practices for TIM Creation

nodes	NodeSetXY	M	Sequence of at least 2 Node entries of XY offsets. A maximum of 63 NodeXY allowed in sequence	Use this field to identify the nodes along the path for the message.
delta	NodeOffsetPointXY	M	Choice of nodes with x and y content. Each entry can range from 20 bits to 32 bits. Each node is computed as an X and Y offset of the prior node unless the prior node is a complete latitude-longitude entry (64 bits).	
attributes	NodeAttributeSetXY	O	Only use if attributes are needed such as changes to lane width and elevation	
computed	ComputedLane	M	Use this if translating the data defined by another lane	
referenceLaneID	LaneID	M	Set the integer value to identify a road lane within an intersection. The left-most lane in the direction of traffic is recommended as the reference lane. Valid values: 0 to 254 Set to 0 when the lane ID is unknown.	
offsetXaxis		M	Choice between DrivenLineOffsetSm or DrivenLineOffsetLg. DrivenLineOffsetSm is an integer ranging from -2047 up to 2047 to indicate the offset in centimeters from a computed lane. DrivenLineOffsetLg is an integer ranging from -32767 up to 32767 to	

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Intelligent Transportation Systems Joint Program Office

			indicate the offset in centimeters from a computed lane. Use when the offset exceeds +/- 2047cm. These values represent values in centimeters from east or west.	
offsetYaxis		M	Choice between DrivenLineOffsetSm or DrivenLineOffsetLg Refer to guidance for offsetXaxis for how to set this value. These offsets represent values in centimeters from north to south.	
rotateXY	Angle	O	Only use if the computed lane is oriented at a different angle than the reference lane. Valid values: 0 to 28800 with each bit representing 0.0125 degrees. The value represents a range from 0 to 359.9875 degrees.	
scaleXaxis	Scale-B12	O	Do not use.	Ignore
scaleYaxis	Scale-B12	O	Do not use.	Ignore
regional	RegionalExtension	O	Do not use.	Ignore

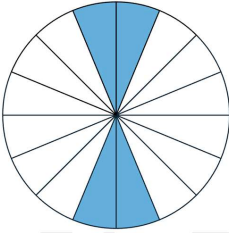
2.7 GeometricProjection Implementation Guidance

GeometricProjection is used to define a circular geofence.

Table 7 Guidance for ASN.1 Entities Contained within GeometricProjection within the GeographicalPath dataframe

Name	ASN.1 Entity (J2735)	Mandatory (M)/	TIM Creation	TIM Interpretation
------	----------------------	----------------	--------------	--------------------

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		Optional (O)		
direction	Heading Slice	M	 • This field is used to define a bit string that represents a unit circle with 16 slices of 22.5 degrees going clockwise. When a slice is set to a value of 1, it represents the direction of travel that the TIM applies to. • The center of the slice range must be the tangent to the roadway at the start location of the sign. • The direction is based on a compass • Ex: a bit string value of 0x8181 would be represented by the shaded unit circle above. This indicates an allowed direction of travel due north or due south at 45-degree angle. • If there are no heading restrictions and the TIM applies in all directions set to all ones (1)	• If the bit string is set to all 1s, the TIM applies in all directions of travel
extent	Extent	O	• Do not use.	• Ignore

laneWidth	LandWidth	O	Do not use.	Ignore
circle	Circle	M	Defines a circle centered at a point and extended to the defined radius.	
center	Position3D	M	- Set latitude, longitude and elevation of the center of circle. - based on WGS-84	Interpret the latitude, longitude and elevation as location of the center of the circle geofence
radius	Radius-B12	M	- Set integer value from 1 to 4094 representing radius from center of circle in the specified units. - Value 4095 is used for unknown	The units for the radius of the circle is defined by the units data element
units	DistanceUnits	M	- Choose from unit list below: 0 – centimeters 1 – centimeters in steps of 2.5 2 – decimeter 3- meter 4 – kilometer 5 – foot (US foot) 6 – yard 7 – mile (US mile) - Recommended to use metric units	Use this field to identify units for the circle geofence
regional	RegionalExtension	O	Do not use.	Ignore

3 Additional Guidance

3.1 Priority

During the Wyoming Interoperability Test Event, it was observed that none of the tested OBUs utilize the ASN.1 entity `signPriority`, derived from J2735, to prescribe a display hierarchy for received TIMs. The intended purpose of `signPriority` is to allow vendors to ensure that higher priority TIMs take precedence over less critical ones. Although no participants implemented `signPriority`, some vendors applied their own prioritization logic:

- One vendor prioritized Work Zone (WZ) TIMs over Weather (W) TIMs, so only the WZ TIMs displayed when both were active in the same area.
- The same vendor also prioritized the slowest active Speed Limit (SL) TIM when multiple SL TIMs were active in an area simultaneously.
- Another vendor's OBUs experienced constant flickering and switching between two active SL TIMs regardless of priority.

Priority values of TIMs will be determined by vendors and deployers. Based on the results from the Wyoming Interoperability Test Event, it is suggested that TIMs with the highest `signPriority` value be displayed when there is more than one Tim relevant in a particular geofence. A sample categorization scheme for TIM priority levels is defined below. The most life-threatening situations requiring immediate action are grouped at Priority 7. High-impact safety events with slightly more reaction time are at Priority 6, operational/regulatory messages are at Priority 4, and convenience information and minimal-impact items are at Priority 2. Priority 0 indicates that a priority level was not defined the TIM.

Priority 7 (Emergency or Critical Alerts)

- Emergency Vehicle Alert: Requires immediate driver action to clear paths for emergency response vehicles, directly impacting life-saving operations.
- Wrong-way Driver Warning: Presents imminent head-on collision risk requiring immediate evasive action.
- Immediate Hazard Warnings (debris, animal, pedestrian in roadway): Requires split-second reaction to avoid collision.

Priority 6 (High Priority)

- Crash/Collision Ahead: Active incident scene with potential for secondary crashes if drivers don't slow/stop appropriately.
- Hazardous Road Conditions (black ice, oil spill): Conditions that dramatically alter safe vehicle handling with little visual indication.
- Railroad Crossing Warning: Time-critical information to prevent catastrophic train-vehicle collisions.
- Work Zone Warnings (especially with workers present): Combines changed road conditions with human safety risk.

- Severe Weather Advisories (flash flooding, tornado): Rapidly changing conditions that may require immediate driver response.
- Road Closure Ahead: Prevents drivers from becoming stranded or entering unsafe areas.

Priority 4 (Medium Priority)

- Lane Closure Information: Requires merging action but typically with more advance notice than higher categories.
- Reduced Speed Zone: Important for safety but allows more reaction time than immediate hazards.
- Bridge Height/Width Restrictions: Critical for specific vehicles but not universal safety impact.
- Traffic Congestion/Slowdown Warning: Reduces rear-end collision risk but generally provides adequate stopping distance.
- Detour Information: Important for traffic flow but secondary to immediate safety concerns.
- Changing Traffic Patterns: Requires attention but typically with adequate warning signage.

Priority 2 (Low priority)

- Travel Time Information: Primarily convenience-focused with minimal direct safety implications.
- HOV/Express Lane Status: Regulatory information with limited safety impact.
- Parking Availability Information: Convenience information with negligible safety impact.
- Rest Area Status: Non-critical travel planning information.

Priority 0

- Priority not defined for the TIM

For TIMs sharing the same priority type, the following criteria can be useful for determining which TIMs take precedence:

- Proximity to driver
- Severity within type
- Recency of information
- Traffic volume affected
- Expected duration
- Alternative routes available

This prioritization practice is considered a best practice because TIMs function as digital traffic control devices similar to Changeable Message Signs (CMS). Prioritization is intended to address the digital version of “sign clutter” as described in **MUTCD Section 2A.20** (Paragraph 2), ensuring drivers receive the most relevant information when there are multiple competing messages.

3.1.1 Relevant MUTCD Guidance

The MUTCD (11th Edition) Section 2A.13 (Paragraphs 12-13) on Standardization of Sign Location provides the following guidance (emphasis added):

Section 2A.13 – Standardization of Sign Location

- Support:
 - Priority is critical where space is limited and multiple signs are needed.
 - Overloading drivers with too much information is undesirable.
 - Priority depends on site conditions (e.g., guide signs and warnings near exit ramps may take precedence over regulatory signs).
- Guidance:
 - Regulatory and warning signs are generally more critical than guide signs.
 - Guide signs should be relocated if conflicts occur.
 - Minimize sign clutter; less-critical signs should be moved or omitted.

Application of Section 2A.13 to TIMs:

TIM broadcasts should be limited in duration, location, and recurrence, focusing on real-time, actionable information.

Section 2A.20 – Excessive Use of Signs

- Guidance:
 - Use signs judiciously to maintain effectiveness and avoid clutter.
 - Regulatory and warning signs should be used conservatively.
- Support:
 - Sign clutter occurs when too many signs compete for attention, reducing effectiveness.
 - Traffic control devices should provide only necessary information for safe and efficient operation.
 - Existing signs should be reviewed periodically to reduce clutter.

Application of Section 2A.20 to TIMs:

Avoid excessive or redundant TIM broadcasts. Messages should be relevant and necessary for driver decision-making.

Section 4F.18 – Preemption and Priority Control

- Traffic signals responding to multiple vehicle classes should prioritize based on importance and stopping difficulty (e.g., train > emergency vehicles > transit).

Application of Section 4F.18 to TIMs:

Similar prioritization principles should apply to TIMs, ensuring critical messages override less important ones.

3.2 Geofence

Under the “Connecting the West” (CTW) project, TIM definition guidance was developed for specific scenarios ranging from high wind gust alerts, regulatory reduced speed, road construction, and traffic

congestion ahead. The CTW TIM guidance provides conditions that lead to the generation of the specified alerts, suggested ITIS phrases and codes, geofence construction, and the timing of a given TIM.

The geofence defines the area of relevance for a TIM and determines the locational boundary in which the message will be presented or displayed to vehicles on the roadway. The CTW guidance defines 5 key points for creating various TIM geofences. These points determine the shape, length and direction of the geofence. Below is a generalization for defining the key points in a TIM geofence.

Point A: Beginning of the area where the incident/restriction occurs

Point B: End of the area where the incident/restriction occurs

Point C: Start of the TIM geofence

Point D: End of the TIM geofence

Point E: The width of the TIM geofence

The CTW project also provided the following guidance on creating TIM geofences when there is not a need for upstream or downstream alerts:

Start to End Extent

The TIM geofence must cover the entire length of the roadway to which it is related. The start and end points of the geofence (Points C and D) should match the beginning and ending (Points A and B) of the road section with the vehicle restriction.

Width Extent

The defined width of the geofence should cover the entire roadway (both directions, where applicable). The width should not extend significantly beyond the edge of the roadway.

Opposite Direction Consideration

For a divided highway, a separate TIM should be defined for each roadway direction, as each extent may differ. In the case of a two-lane, two-way road, conditions that restrict trailers are assumed to impact both directions. Therefore, a single TIM with a geofence extending across both directions of travel can be created. This single TIM would include a directionality that covers both directions.

There are some scenarios when the alert should be sent ahead of time before approaching the area where the incident/restriction occurs. This may be the case when warning drivers of congestion, an accident or collision, or an active work zone ahead. In these cases, the start and/or end points of the geofence (Points C and D) need to be set several feet before the point A and B depending on the speed limit in the area.

3.3 Timing

The timing conditions for a TIM vary based on the type of alert being sent. The start time should be set as accurately as possible. If the event is planned such as with construction, then the planned start date and time should be used. For most other events, the start time should be set as soon as the incident is detected. Below are four scenarios taken from the CTW TIM guidance to provide additional details for setting the duration time.

Uncertain end time/permanent restrictions:

The exact duration of unplanned incidents is typically unknown. To handle this uncertainty, TIMs may be created with indefinite end times, however close attention must be paid to the situation so that the TIM can be removed as soon as it is no longer relevant. If the clearance time for an incident can be reasonably estimated, the incident ahead TIM should have an end-time based on this estimated clearance

time. Creating a TIM with a defined end-time is preferred over an indefinite end-time, since a defined end-time mitigates the possibility of drivers receiving incorrect, stale information. In either case, defined or indefinite end-time, careful monitoring of the incident situation is required, and associated TIMs updated with a precise end-time as soon as such end-time can be determined with confidence.

Involving road workers:

To maximize the effectiveness of the "look-out-for-workers" TIM in enhancing driver caution and speed limit compliance, it is crucial to ensure the highest level of message accuracy. The TIM should be available to drivers only when workers are actively present and should end as soon as workers leave the site. For example, if workers are not present during nighttime hours at a work site, then the TIM should not be active at those times. When creating the TIM, it is recommended to include an estimated end time and update it based on real-time field conditions.

Planned workzone/construction:

The timing of this road construction ahead warning TIM is completely dependent upon the timing of the related road construction zone. When the work zone is put in place this warning TIM should be activated and remain active for the entire duration that the work zone is in place, deactivating only once the work zone is removed. Given that most road construction is planned, it may be possible to use the planned start and end dates as the timing for the TIM activation. However, for most road construction projects the planned start and end dates are estimates, and to achieve the necessary timing accuracy for in-vehicle messaging, the TIM start and end times will have to be updated based on real-time field conditions.

Rapidly expanding congestion:

In the case of rapidly expanding congestion zones, regular timed updates to the congestion warning TIM may be required to keep the TIM geofence extent valid (see Geofence Extent discussion). If regular timed updates are made for a given TIM, each update may have a short, timed duration matching the TIM update rate. For example, if a TIM is updated every 15 minutes, then each update version of the TIM may have start and end times defined to roughly match the 15-minute update interval. Handling TIM update intervals shorter than 30 minutes may require traffic monitoring systems able to automatically identify congestion build up on the roadway and automatically determine the congestion zone boundaries necessary to generate each TIM update.

4 Example TIM Message

Below is an example of a speed limit TIM utilizing guidance above. It includes both a circle and a path geographic region.

```
"tim": {
  "msgCnt": "1",
  "timeStamp": "2024-10-01T10:00:00.0Z",
  "packetID": "8D44FF260FFFD00B01",
  "urlB": "null",
  "dataFrames": [
    {
      "doNotUse1": "0",
      "frameType": "2",
      "msgId": {
        "furtherInfoID": "0000000000000000"
      },
      "startYear": "2024",
      "startTime": "401760",
      "durationTime": "8640",
      "priority": "4",
      "doNotUse2": "0",
      "regions": [
        {
          "anchor": {
            "latitude": "41153593",
            "longitude": "-104655785",
            "elevation": "-4096"
          },
          "laneWidth": "5000",
          "description": "geometry"
        },
        {
          "geometry": {
            "direction": "1111111111111111",
            "circle": {
              "center": {
                "lat": "41153593",
                "long": "-104655785"
              },
              "radius": "1",
```

```

        "units": "7"
      }
    },
    {
      "anchor": {
        "latitude": "41153593",
        "longitude": "-104655785",
        "elevation": "-4096"
      },
      "laneWidth": "50",
      "directionality": "3",
      "closedPath": "false",
      "description": "path",
      "path": {
        "offset": "11"
        "ll": "nodes"
        "nodes": [
          {
            "NodeSetLL": {
              "delta": "node-LatLon" {
                "lon": "-00005853",
                "lat": "00002057"
              }
            },
            "NodeSetLL": {
              "delta": "node-LatLon" {
                "lon": "-00008454",
                "lat": "00001861"
              }
            },
            "NodeSetLL": {
              "delta": "node-LatLon" {
                "lon": "-0.0009494",
                "lat": "0.0001469"
              }
            },
            "NodeSetLL": {
              "delta": "node-LatLon" {
                "lon": "-0.0011835",
                "lat": "0.0001175"
              }
            },
            "NodeSetLL": {
              "delta": "node-LatLon" {

```

Example TIM Message

```
        "lon": "-0.0010405",
        "lat": "0.0000490"
    },
    "NodeSetLL": {
        "delta": "node-LatLon" {
            "lon": "-0.0007283",
            "lat": "0.0000196"
        }
    },
    "NodeSetLL": {
        "delta": "node-LatLon" {
            "lon": "-0.0007413",
            "lat": "0.0000000"
        }
    }
}
]
"scale": "0"
}
}
]
"doNotUse3": "0",
"doNotUse4": "0",
"content": "advisory"
"advisory": [
    {
        "item": "itis"
        "itis": "268"
    },
    {
        "item": "itis"
        "itis": "12599"
    };
    {
        "item": "itis"
        "itis": "8720"
    }
]
"contentNew": "0"
}
]
}
```

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